1 network foundamentals & delays

Problem 1.1

file 5mb transferred over 3 link in series, single packet

capacity $R = 10$ Mb/s

length $d = 1500$ km

Propagation speed $V = 2 \times 10^8 \frac{m}{s}$

a) transmission delay on 1 link

$$d_{trans} = \frac{L}{R} = \frac{5mB}{10 \frac{mb}{s}} = \frac{40 mb}{10 \frac{mb}{s}} = 4 \text{ seconds on 1 link}$$

5 bytes → 5 · 8 bits

40 mb

b) propagation delay on 1 link

$$d_{prop} = \frac{d}{v} = \frac{1500 \text{ km}}{2 \times 10^8 \frac{m}{s}} = 7.5 \times 10^{-3} s$$

$1500 \text{ km} = 1.5 \times 10^6 \text{ m}$

c) total end to end delay

$$d_{total} = N \cdot d_{trans} + N \cdot d_{prop}$$

N → links

$$= 3 \cdot 4 + 3 \cdot 7.5 \times 10^{-3}$$

$$= 12 + 22.5 \times 10^{-3} \rightarrow 0.0225$$

$$= 12.0225 \text{ seconds}$$

d) which delay dominates, why?

transmission because a lot larger than propagation

Problem 1.2

a) list the five layers of internet stack protocol, top to bottom

application

Transport

network

link

physical

b) a web browser sends a HTTP GET request

at which layer does each of the following operations occur

i) breaking request into segment

TCP/UDP

ii) adding source / des IP address

link

iii) explain concept of encapsulation : what happens to data as it travels down stack

add other data